Chair travel case

ceremonial chairs · build plans · Rev L (12 mm CNC-cut kit, ASA printed fittings)

Case outside	1024 W × 1014 D × 1764 H mm on castors
Inside	1000 × 990 × 1605 mm, lined 20 mm EPE, 20 mm EVA floor
Chair	927 × 920 × 1547 (maker's STEP, 9 Nov), 15–18 mm air gap all round
Build	12 mm black hexa ply walls, top and door on a 17 mm formply base; CNC-cut panels, glued + pinned, 30 mm single angle, 260 mm deep hinged door (five lift-off hinges) on a 12 mm hybrid edge split; ASA printed handles, foot rails and brackets
Access	3 butterfly latches; the door swings open on the left and a strap stops it at 95 degrees (needs 770 mm clear in front); lift it off its hinges to remove it; chair slides out forward over a flush threshold
Mobility	100 mm castors (100 kg each, 2 braked) bolted to the base at the corners, 6 handles
Weight	about 98 kg empty, 151 kg packed (calculated: weigh the first one)

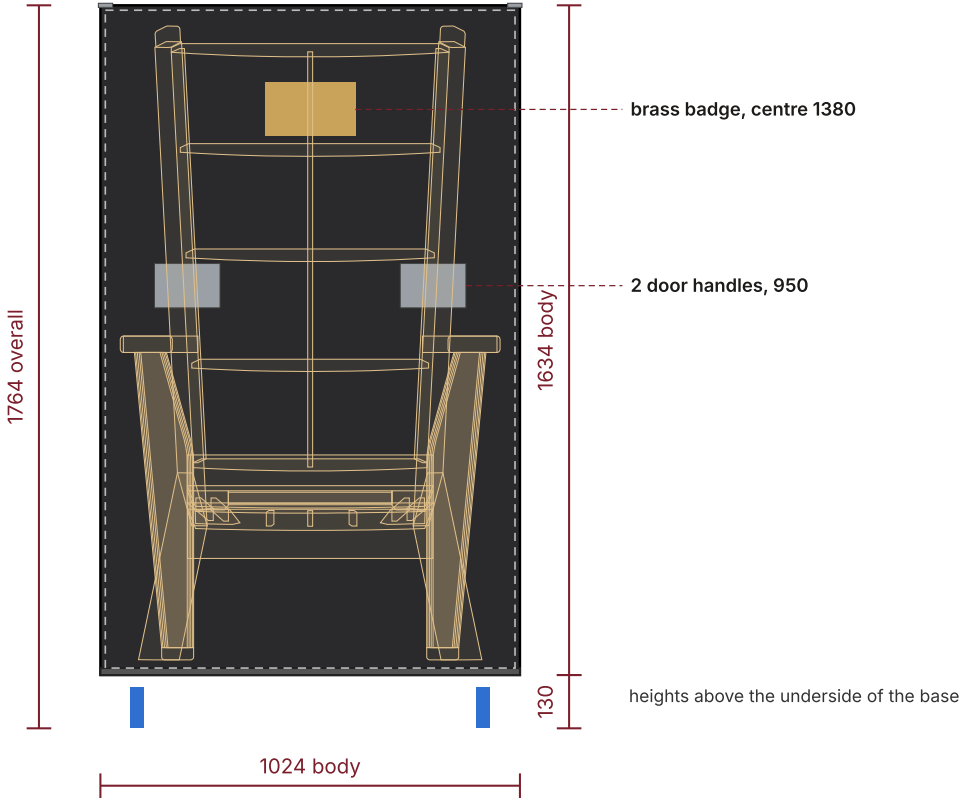
Before the final CNC files go: confirm the real ply thickness, and mate the hybrid joiner on two offcuts to measure its gap (drawn as 8 mm). Set the foot rails on the first finished chair. Print and test one handle first. Chair geometry is maker's STEP model. Badge needs the client' official vector logo.



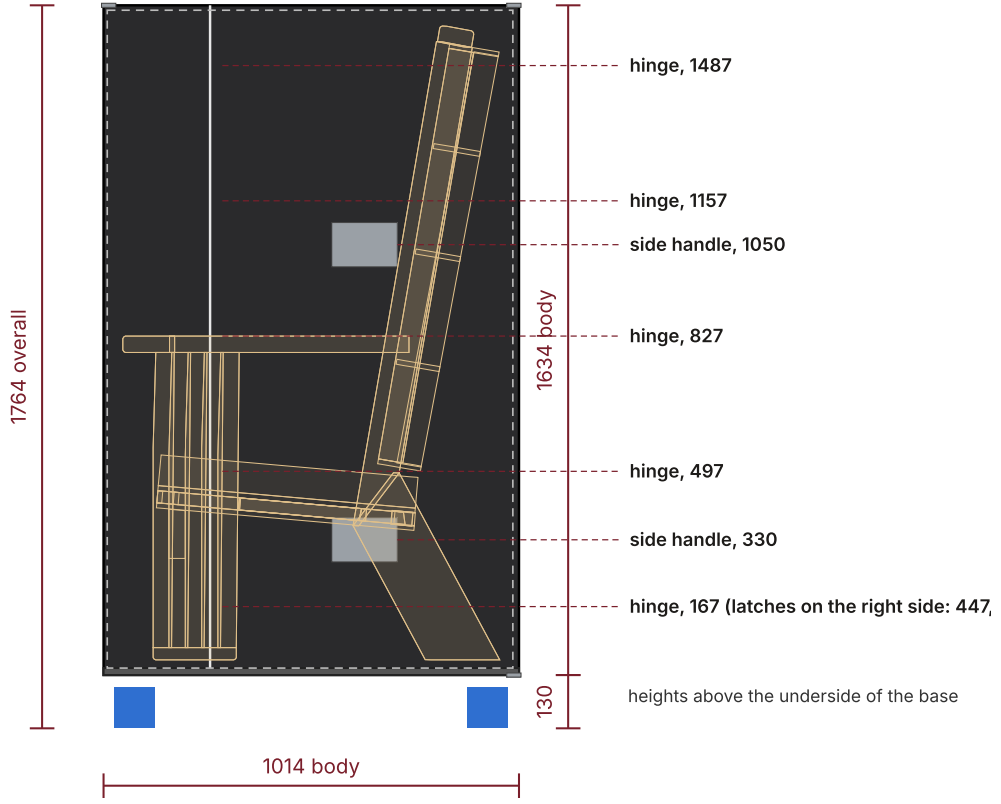


Elevations

FRONT (door shut), chair outlined

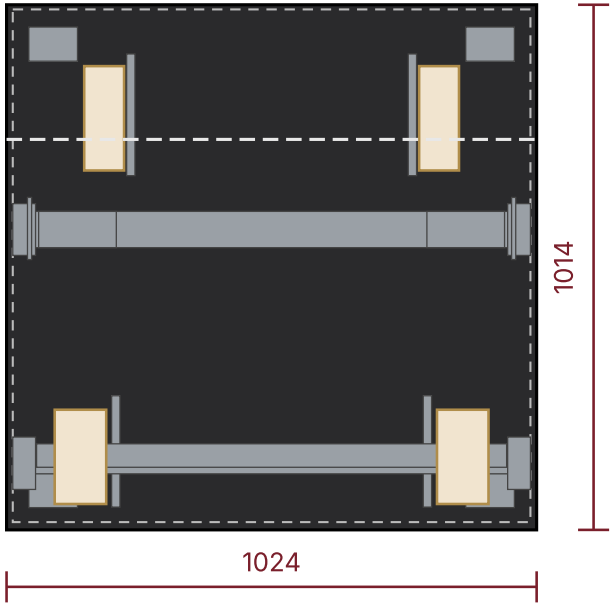


LEFT SIDE, door at left (split 260 back), hinges this side

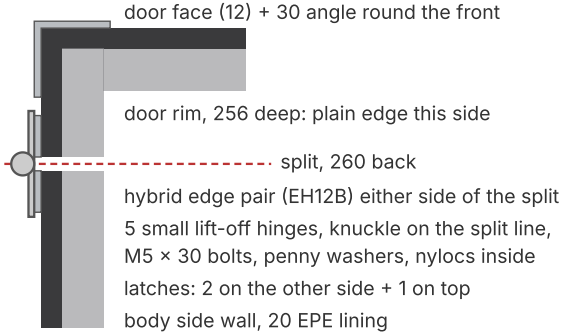


Plan and details

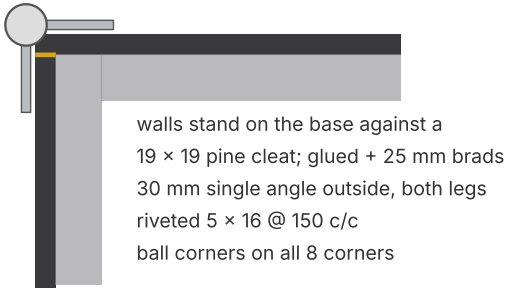
PLAN at floor, door at top (hinged on the left)



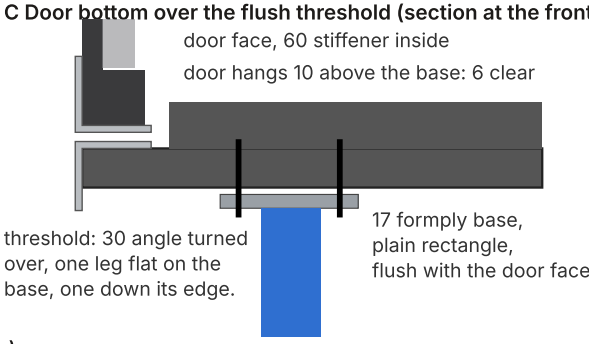
A Door and split (plan section, hinge side)



B Rear corner (plan section)



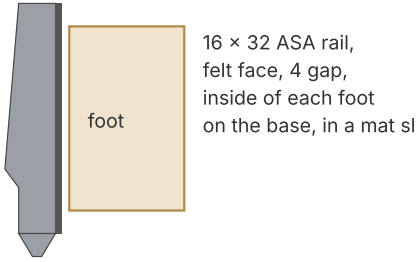
C Door bottom over the flush threshold (section at the front)



Nothing stands up in the doorway.

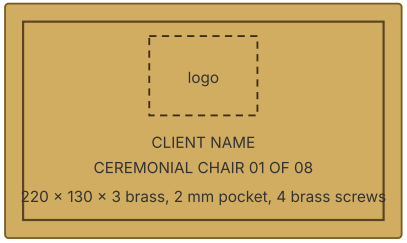
D Foot rail, ASA print (plan)

back rib stops the chair, not these

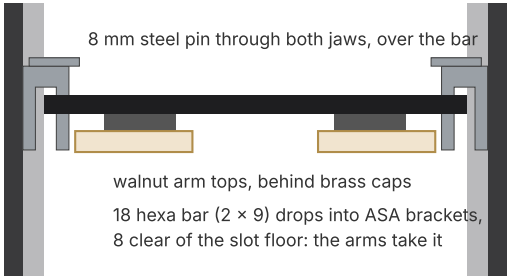


flared end towards the door

E Brass badge

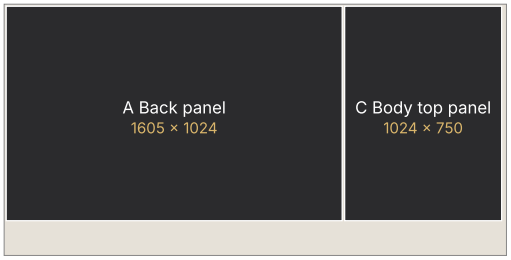


F Hold-down bar (section across case)

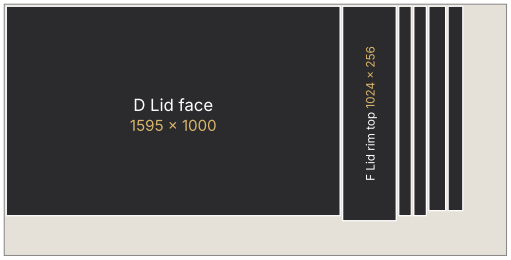


Cut list · hexa ply, 4 sheets per case

Sheet 1



Sheet 2



Sheet 3



Sheet 4



CNC parts, one case

Part	Name	Size	Qty
A	Back panel	1024 × 1605	1
B-L	Body side panel LEFT (hinge side: mirror of B-R without the latch notches)	738 × 1605	1
B-R	Body side panel RIGHT	738 × 1605	1
C	Body top panel	1024 × 750	1
D	Lid face	1000 × 1595	1
E-L	Lid rim side LEFT (hinge side: no notches)	256 × 1595	1
E-R	Lid rim side RIGHT	256 × 1595	1
F	Lid rim top	1024 × 256	1
G	Lid bottom stiffener	1000 × 60	2
H	Back rib	976 × 80	1
J	Hold-down bar	976 × 70	1
M	Base	1024 × 1014	1

Panels are CNC cut by the supplier from the Rev L DXF pack (12 mm), with handle openings, bracket bolt holes, latch notches, the badge pocket and castor marks. Left and right are mirror images. The hybrid joiner gap is drawn as 8 mm and the latch notches as 49 (body) and 23 (door) deep: best estimates from the suppliers' published sizes. Extrusion runs: 30 mm single angle 1024, 264, 264, 964, 720, 226, 1574, 695, 1577, 201, 720, 226, 1574, 695, 1577, 201, 964, 964, 964; hybrid joiner 1578, 1578, 974, 1551, 1551, 974.

Build order

1. **Order and collect.** Panels: order the Rev M (12 mm, hinged door) CNC-cut kits from the chosen panel supplier. Before the final files go, confirm the real sheet thickness and buy one 3 m length of Armor EH12B hybrid edge, one latch and five Titan small lift-off hinges (RC-SP-LH12, 60 × 50) so the split gap, the latch notches and the hinge seating can be checked. Extrusion, latches, hinges, ball corners, rivets and foam: Titan AV (ask for uncut 3 m extrusion lengths: five verticals are over 1.5 m). Bolts, screws, glue, contact spray and paint from Bunnings. Start the ASA prints (list on the ASA prints tab) while the panels are being cut.
2. **Print the ASA parts.** Print one set per case in black ASA on the Bambu printers, enclosure closed: 6 recessed handles (face down), 2 front and 2 rear foot rails, 2 rib brackets, 2 bar brackets, 2 front bumpers, 1 label holder, 1 door landing pad. Plus one rivet drill guide for the workshop. All parts fit a 256 mm bed. Every load-carrying part (handles, brackets, rails, bumpers) prints with 5 walls and 40% infill, no supports; the 4 mm walls of the handle come out solid. Print one handle first, bolt it into an offcut with a 130 × 75 cut-out and hang your full weight off it before running the rest. If it creaks, flexes or whitens, stop.
3. **Check the kit.** Lay out one case's parts against the parts list (A to M). Check the hexa face is on the outside of both side panels and both door rims: left and right are mirror images, and only the RIGHT ones (latch side) have latch notches; the left ones (hinge side) are plain. The door face (D) and rims (E) are 1595 high, 10 mm shorter than the body walls: the door hangs 10 mm above the base and never touches it. The base (M) is a plain rectangle. Offer a handle into its cut-out and a latch up to its notches. Mate two EH12B hybrid edge halves on 12 mm offcuts and measure the gap between the two ply edges: the kit is drawn for 8 mm and works from 6 up to about 12 (a bigger gap just sets the door that much further forward). Lay the test hinge across the mated pair, knuckle on the seam: both leaves must sit flat. If a leaf overhangs the edge of the hybrid leg, cut 2 mm aluminium packers for under it.
4. **Prep the parts.** Paint the cut edges of the base black. Round over the edges of the back rib (H) and the hold-down bar (J): each is a single 12 mm strip. Glue 20 mm EVA to the front face of the rib and two EVA pads under the bar where it sits on the arms. Cut the three 19 × 19 pine cleats.
5. **Glue up the body and the door.** Body: glue and screw the pine cleats to the base, 12 mm in from the back and side edges (the wall thickness). Stand the back (A) and the two sides (B-L plain edge to the front on the left, B-R notched edge to the front on the right) on the base against the cleats; glue (PU) and pin with 25 mm brads, then screw through each wall into its cleat every 200 mm with 8g x 25 screws, 6 mm above the base: the 13 mm upper leg of the base angle covers those heads. Glue and pin the top (C) on, notch to the front. Door: build it ON the body so the two halves cure as one box. Lay two 10 mm packers on the base in the doorway (the door's bottom edge sits 10 mm above the base) and stand packers on the split edges of the body as thick as the gap you measured in step 3 (about 8 mm). Glue and pin the rims (E-L plain on the left, E-R notched on the right) to the edges of the face (D), the top strip (F) over all three, and the two 60 mm stiffener strips (G), glued face to face inside the face, flush with its bottom edge. Tape across the split so no glue bridges it. Check the diagonals of the door face are within 2 mm. Leave it overnight, then lift the door away.
6. **Fit the split frame.** Measure the built box and cut the extrusions to it, not to the drawing. Fit the EH12B hybrid edge pair up both sides and across the top of the split, door half on the door's bottom corners: a ball would hang below the door and hit the base. No flat corners or brace plates.
8. **Latches, hinges and handles.** Hang the door in this order. 1: stand the door in the doorway on two 6 mm packers laid on the threshold under its bottom side angles, tongues in their grooves. 2: fit the three butterfly latches into the pre-cut notches (two on the right, one on top; dish half on the body, catch half on the door) and latch the door shut: the hybrid edges and latches now hold it exactly where it has to hang. 3: lay each lift-off hinge closed across the left split, knuckle on the seam, at the five marks, all five the same way up so the door can lift off (pin leaf on the body with the pin pointing up, or pin leaf on the door pointing down). Drill 5.5 mm through each hinge hole (four per hinge) and bolt with M5 × 30 button-head bolts, penny washers and nylocs inside. 4: pull the packers, open the door and check it swings free and drops back onto its tongues without lifting. 5: screw the ASA landing pad onto the threshold return on the right, 60 mm back from the front, ramp to the front (2 × 8g x 20 countersunk): the door's bottom angle should just kiss it as the door shuts. 6: bolt the check strap (25 mm webbing, 385 mm between holes) through the top panel 40 mm behind the split and the door's top rim 40 mm in front of it, 300 mm from the hinge side, M5 bolts with penny washers; it must stop the door just past square. Then the six ASA recessed handles into the 130 × 75 cut-outs (two each side, two on the door face): 6 x M5 × 25 black button-head bolts each, washers and nylocs inside. Bolt the two front bumpers to the inside of the door face (4 x M5 × 25 each, holes pre-drilled) but leave their EVA pads off until the first-chair fit. Rivet the label holder to the back.
9. **Castors.** The base has the four castor plate centres marked (90 mm in from the sides, 76 mm in from the front and back). Centre each plate on its mark, drill through its holes, and bolt with M8 × 40 cup-head bolts, heads inside (they end up under the floor mat), nylocs under the plate. Braked pair at the door end.
10. **Brackets for the rib and bar.** The side walls come with the bracket bolt holes pre-drilled. Bolt the two ASA rib brackets (low, near the back, leaning back 22 degrees: the holes set the angle) and the two ASA bar brackets (high, near the front) to the walls with 4 x M5 × 30 button-head bolts each, heads outside. Tie an 8 mm steel hitch pin (about 115 long, with its R-clip) to each bar bracket on a short lanyard, and check it slides through both jaws.
11. **Line it.** Cut the foam to the built box. Spray-contact 20 mm EPE on the back, sides and top of the body and on the door face and the door's hinge-side rim and top rim; use 10 mm EPE on the door's latch-side rim (that rim swings in an arc past the chair). Start the door rim linings 25 mm above the door's bottom edge. Cut pockets for the hinge nuts, brackets, bumpers, latch pans and handle pockets. Then the 20 mm EVA floor mat between the linings, with a 30 × 222 notch cut out of its front right corner (the door swings in over that corner) and the four foot-rail slots (16 wide; mark them from the first chair, step 12, before gluing the mat down). Stick the brush strip under the door's bottom stiffener.
12. **First-chair fit.** Drop the rib into its brackets. With the door open against its strap and the castors braked, lift the first chair in from the front and slide it back until its rear legs touch the rib foam, centred side to side. Mark the inside edge of each foot on the mat. Lift the chair out, cut a 16 mm slot in the mat 4 mm inside each mark, and screw the four ASA foot rails straight onto the base in the slots (5 × 8g x 25 each), flared end towards the door, felt on the foot side. Roll the chair back in. Lay the bar into its brackets on the arm tops behind the brass caps, push a steel pin through each bracket above it and clip both pins. Now fit